Solutions by:
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Calculus for Biology and Medicine, 3rd Edition, Claudia Neuhauser

10.4 # 1, 7, 11, 14, 17, 20, 25, 29, 36, 43, 44, 46

1) $f(x,y)=2x^3+y^2$; $(1,2,6)$

$$z-z_0=\frac{\partial f}{\partial x}(x_0,y_0)(x-x_0)+\frac{\partial f}{\partial y}(x_0,y_0)(y-y_0)$$

$$\frac{\partial f}{\partial x}=6x^2 \quad \frac{\partial f}{\partial x}(1,2)=6(1)^2=6$$

$$\frac{\partial f}{\partial y}=2y \quad \frac{\partial f}{\partial y}(1,2)=2(2)=4$$

$$\boxed{z-6=6(x-1)+4(y-2)}$$

7) $f(x,y)=e^{x^2+y^2}$; $(1,0,e)$

$$z-z_0=\frac{\partial f}{\partial x}(x_0,y_0)(x-x_0)+\frac{\partial f}{\partial y}(x_0,y_0)(y-y_0)$$

$$\frac{\partial f}{\partial x}=2xe^{x^2+y^2} \quad \frac{\partial f}{\partial x}(1,0)=2(1)e^{(1)^2+(0)^2}=2e$$

$$\frac{\partial f}{\partial y}=2ye^{x^2+y^2} \quad \frac{\partial f}{\partial y}(1,0)=2(0)e^{(1)^2+(0)^2}=0$$

$$\boxed{z-e=2e(x-1)+0(y-0)}$$

11) $f(x,y)=y^2x+x^2y$; $(1,1)$

$$\frac{\partial f}{\partial x}=y^2+2xy \quad \frac{\partial f}{\partial x}(1,1)=(1)^2+2(1)(1)=3 \checkmark$$

$$\frac{\partial f}{\partial y}=2xy+x^2 \quad \frac{\partial f}{\partial y}(1,1)=2(1)(1)+(1)^2=3 \checkmark$$

$$\lim_{(x,y)\to(1,1)} y^2x+x^2y=(1)^2(1)+(1)^2(1)=2 \checkmark$$

$$\boxed{\frac{\partial f}{\partial x},\frac{\partial f}{\partial y}, \text{ and } \lim_{(x,y)\to(1,1)} f(x,y) \text{ exist, therefore } f(x,y) \text{ is differentiable at } (1,1).}$$

14) $f(x,y)=e^{x-y}$; $(0,0)$

$$\frac{\partial f}{\partial x}=e^{x-y} \quad \frac{\partial f}{\partial x}(0,0)=e^{0-0}=1 \checkmark$$

$$\frac{\partial f}{\partial y}=-e^{x-y} \quad \frac{\partial f}{\partial y}(0,0)=-e^{0-0}=-1 \checkmark$$

$$\lim_{(x,y)\to(0,0)} e^{x-y}=e^{0-0}=1 \checkmark$$

$$\boxed{\frac{\partial f}{\partial x},\frac{\partial f}{\partial y}, \text{ and } \lim_{(x,y)\to(0,0)} f(x,y) \text{ exist, therefore } f(x,y) \text{ is differentiable at } (0,0).}$$

17) $f(x,y)=x-3y$; $(3,1)$

$$L(x,y)=f(x_0,y_0)+\frac{\partial f}{\partial x}(x_0,y_0)(x-x_0)+\frac{\partial f}{\partial y}(x_0,y_0)(y-y_0)$$

$$\frac{\partial f}{\partial x}=1 \quad \frac{\partial f}{\partial x}(3,1)=1$$

$$\frac{\partial f}{\partial y}=-3 \quad \frac{\partial f}{\partial y}(3,1)=-3$$

$$f(3,1)=(3)-3(1)=0$$

$$\boxed{L(x,y)=0+1(x-3)-3(y-1)}$$

20) $f(x,y)=\cos(x^2y)$; $(\frac{\pi}{2},0)$

$$L(x,y)=f(x_0,y_0)+\frac{\partial f}{\partial x}(x_0,y_0)(x-x_0)+\frac{\partial f}{\partial y}(x_0,y_0)(y-y_0)$$

$$\frac{\partial f}{\partial x}=-2xy\sin(x^2y)$$

$$\frac{\partial f}{\partial y}=-x^2\sin(x^2y)$$

$$f(\tfrac{\pi}{2},0)=\cos\left(\left(\tfrac{\pi}{2}\right)^2(0)\right)=1$$

$$\frac{\partial f}{\partial x}(\tfrac{\pi}{2},0)=-2(\tfrac{\pi}{2})(0)\sin\left((\tfrac{\pi}{2})^2(0)\right)=0$$

$$\frac{\partial f}{\partial y}(\tfrac{\pi}{2},0)=-2(\tfrac{\pi}{2})^2\sin\left((\tfrac{\pi}{2})^2(0)\right)=0$$

$$\boxed{L(x,y)=1+0(x-\tfrac{\pi}{2})-0(y-0)}$$

25) $f(x,y)=e^{x+y}$

$$\frac{\partial f}{\partial x}=e^{x+y} \quad \frac{\partial f}{\partial x}(0,0)=e^{(0)+(0)}=1$$

$$\frac{\partial f}{\partial y}=e^{x+y} \quad \frac{\partial f}{\partial y}(0,0)=e^{(0)+(0)}=1$$

$$f(0,0)=e^{(0)+(0)}=1$$

$$\boxed{L(x,y)=1+1(x-0)-1(y-0)}$$

$$L(0.1,0.05)=1+1(0.1-0)+1(0.05-0)$$
$$=1+0.1+0.05$$
$$=\boxed{1.15} \qquad f(0.1,0.05)=\boxed{1.16}$$

29) $f(x,y)=\begin{bmatrix} x+y \\ x^2-y^2 \end{bmatrix}$

$$Df(x,y)=\boxed{\begin{bmatrix} 1 & 1 \\ 2x & -2y \end{bmatrix}}$$

36) $f(x,y)=\begin{bmatrix} \sqrt{x^2+y^2} \\ e^{-x^2} \end{bmatrix}$

$$Df(x,y)=\boxed{\begin{bmatrix} \frac{1}{2}(x^2+y^2)^{-1/2}(2x) & \frac{1}{2}(x^2+y^2)^{-1/2}(2y) \\ -2xe^{-x^2} & 0 \end{bmatrix}}$$

43) $f(x,y)=\begin{bmatrix} x^2-xy \\ 3y^2-1 \end{bmatrix}$; $(1,2)$; $f(1.1,1.9)$

$$Df(x,y)=\begin{bmatrix} 2x-y & -x \\ 0 & 6y \end{bmatrix} \quad Df(1,2)=\begin{bmatrix} 2(1)-(2) & -(1) \\ 0 & 6(2) \end{bmatrix}=\begin{bmatrix} 0 & -1 \\ 0 & 12 \end{bmatrix}$$

$$f(1,2)=\begin{bmatrix} (1)^2-(1)(2) \\ 3(2)^2-1 \end{bmatrix}=\begin{bmatrix} -1 \\ 11 \end{bmatrix}$$

$$\boxed{L(x,y)=\begin{bmatrix} -1 \\ 11 \end{bmatrix}+\begin{bmatrix} 0 & -1 \\ 0 & 12 \end{bmatrix}\begin{bmatrix} x-1 \\ y-2 \end{bmatrix}}$$

$$f(1.1,1.9)=\boxed{\begin{bmatrix} -0.88 \\ 9.83 \end{bmatrix}}$$

$$\begin{aligned} L(1.1,1.9) &= \begin{bmatrix} -1 \\ 11 \end{bmatrix}+\begin{bmatrix} 0 & -1 \\ 0 & 12 \end{bmatrix}\begin{bmatrix} 1.1-1 \\ 1.9-2 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 11 \end{bmatrix}+\begin{bmatrix} 0 & -1 \\ 0 & 12 \end{bmatrix}\begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 11 \end{bmatrix}+\begin{bmatrix} (0)(0.1)+(-1)(-0.1) \\ (0)(0.1)+(12)(-0.1) \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 11 \end{bmatrix}+\begin{bmatrix} 0.1 \\ -1.2 \end{bmatrix} = \boxed{\begin{bmatrix} -0.9 \\ 9.8 \end{bmatrix}} \end{aligned}$$

44) $f(x,y)=\begin{bmatrix} x/y \\ 2xy \end{bmatrix}$; $(-1,1)$; $f(-0.9,1.05)$

$$Df(x,y)=\begin{bmatrix} 1/y & -x/y^2 \\ 2y & 2x \end{bmatrix} \quad Df(-1,1)=\begin{bmatrix} 1/(1) & -(-1)/(1)^2 \\ 2(1) & 2(-1) \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$f(-1,1)=\begin{bmatrix} (-1)/(1) \\ 2(-1)(1) \end{bmatrix}=\begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

$$\boxed{L(x,y)=\begin{bmatrix} -1 \\ 2 \end{bmatrix}+\begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}\begin{bmatrix} x+1 \\ y-1 \end{bmatrix}}$$

$$f(-0.9,1.05)=\boxed{\begin{bmatrix} -0.86 \\ -1.89 \end{bmatrix}}$$

$$\begin{aligned} L(-0.9,1.05) &= \begin{bmatrix} -1 \\ -2 \end{bmatrix}+\begin{bmatrix} 1 & 1 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} -0.9+1 \\ 1.05-1 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -2 \end{bmatrix}+\begin{bmatrix} 1 & 1 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} 0.1 \\ 0.05 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -2 \end{bmatrix}+\begin{bmatrix} (1)(0.1)+(1)(0.05) \\ (2)(0.1)-(2)(0.05) \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -2 \end{bmatrix}+\begin{bmatrix} 0.15 \\ 0.10 \end{bmatrix} = \boxed{\begin{bmatrix} -0.85 \\ -1.90 \end{bmatrix}} \end{aligned}$$

46) $f(x,y)=\begin{bmatrix} \sqrt{2x+y} \\ x-y^2 \end{bmatrix}$; $(1,2)$; $f(1.05,2.05)$

$$Df(x,y)=\begin{bmatrix} \frac{1}{2}(2x+y)^{-1/2}\cdot 2 & \frac{1}{2}(2x+y)^{-1/2} \\ 1 & -2y \end{bmatrix} \quad Df(1,2)=\begin{bmatrix} (2(1)+(2))^{-1/2} & \frac{1}{2}(2(1)+(2))^{1/2} \\ 1 & -2(2) \end{bmatrix}=\begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 1 & -4 \end{bmatrix}$$

$$f(1,2)=\begin{bmatrix} \sqrt{2(1)+(2)} \\ (1)-(2)^2 \end{bmatrix}=\begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$\boxed{L(x,y)=\begin{bmatrix} 2 \\ -3 \end{bmatrix}+\begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 1 & -4 \end{bmatrix}\begin{bmatrix} x-1 \\ y-2 \end{bmatrix}}$$

$$f(1.05,2.05)=\boxed{\begin{bmatrix} 2.0372 \\ -3.1525 \end{bmatrix}}$$

$$\begin{aligned} L(1.05,2.05) &= \begin{bmatrix} 2 \\ -3 \end{bmatrix}+\begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 1 & -4 \end{bmatrix}\begin{bmatrix} 1.05-1 \\ 2.05-2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3 \end{bmatrix}+\begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 1 & -4 \end{bmatrix}\begin{bmatrix} 0.05 \\ 0.05 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3 \end{bmatrix}+\begin{bmatrix} (\frac{1}{2})(0.05)+(\frac{1}{4})(0.05) \\ (1)(0.05)+(-4)(0.05) \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3 \end{bmatrix}+\begin{bmatrix} 0.0375 \\ -0.15 \end{bmatrix} = \boxed{\begin{bmatrix} 2.0375 \\ -3.15 \end{bmatrix}} \end{aligned}$$